

Flying Robots

Quadrotor Dynamic Modeling and Control

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What is a Quadcopter?

A **quadrotor** (quadcopter) is an **unmanned aerial vehicle (UAV)** lifted and propelled by **four rotors** arranged symmetrically.

- First concepts: early 20th century
- Modern revival driven by cheap MEMS sensors and brushless motors
- Widely used in: photography, delivery, search & rescue, research

Key numbers — DJI F450

Mass	1.0 kg
Arm length	225 mm
Max thrust	≈ 20 N
Hover current	≈ 10 A

Why Study Quadcopters?

Academic interest

A quadrotor is a canonical **underactuated**, **nonlinear**, **6-DOF** system — ideal for testing control and estimation algorithms.

Challenge

6 degrees of freedom, only 4 independent actuators $\Rightarrow$ **underactuated**: position and yaw are *not* independently controllable.

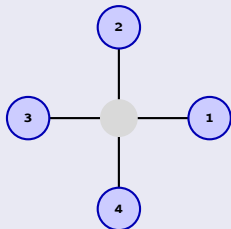
Connections to the course

- Homogeneous transforms & rotation matrices (Ch. 1–2)
- Newton–Euler dynamics (Ch. 4)
- Computed-torque / PD control (Ch. 5)

Rotor Layout and Numbering

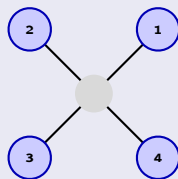
Two common configurations:

“+” Configuration



Rotors on cardinal axes

“×” Configuration



Rotors at 45° — used in practice

Rotation directions: alternating CW/CCW to cancel net yaw torque at hover.

Reference Frames

$\mathcal{W}$ **World (inertial) frame** — fixed, North-East-Down or ENU convention

$\mathcal{B}$ **Body frame** — origin at CoM, x_B toward rotor 1, z_B upward

State variables

Position in $\mathcal{W}$:

$$\xi = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$$

Euler angles (ZYX):

$$\eta = \begin{pmatrix} \phi \\ \theta \\ \psi \end{pmatrix} = \begin{pmatrix} \text{roll} \\ \text{pitch} \\ \text{yaw} \end{pmatrix}$$

Rotation Matrix R

The rotation from body to world uses the **ZYX convention**:

$$R(\phi, \theta, \psi) = R_z(\psi) R_y(\theta) R_x(\phi)$$

Explicit form ($c_\alpha \equiv \cos \alpha$, $s_\alpha \equiv \sin \alpha$):

$$R = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}$$

Gimbal lock

Euler-angle representation has a **singularity** at $\theta = \pm 90$. For aggressive maneuvers, use **unit quaternions** on S^3 .

Angular Velocity Relations

Body angular velocity $\omega^B = (p, q, r)^T$ is related to Euler rates by:

$$\omega^B = W(\eta) \dot{\eta}$$

where the **kinematic matrix** is:

$$W(\eta) = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix}$$

Away from singularity

$$\dot{\eta} = W^{-1}(\eta) \omega^B$$

Near $\theta = 90$

$$\det W \rightarrow 0$$

$\Rightarrow$ use quaternions!

Rotor Thrust and Drag Torque

For each rotor i spinning at angular speed Ω_i :

Thrust force (along z_B)

$$f_i = k_f \Omega_i^2, \quad k_f > 0 \quad [\text{N} \cdot \text{s}^2/\text{rad}^2]$$

Drag (reaction) torque (about z_B)

$$\tau_i^{drag} = k_m \Omega_i^2, \quad k_m > 0 \quad [\text{N} \cdot \text{m} \cdot \text{s}^2/\text{rad}^2]$$

k_f Thrust coefficient — depends on blade geometry, air density

k_m Drag coefficient — ratio k_m/k_f typically 10^{-2} – 10^{-3}

Rotor Speed $\rightarrow$ Virtual Inputs

Define $\Gamma_i = \Omega_i^2$. The **four virtual inputs** are total thrust f and three torques:

$$\begin{pmatrix} f \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{pmatrix} = \underbrace{\begin{bmatrix} k_f & k_f & k_f & k_f \\ 0 & lk_f & 0 & -lk_f \\ -lk_f & 0 & lk_f & 0 \\ k_m & -k_m & k_m & -k_m \end{bmatrix}}_{\Gamma \text{ (allocation matrix)}} \begin{pmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{pmatrix}$$

l = arm length. Γ is **invertible** $\Rightarrow$ unique mapping from $u = (f, \tau_\phi, \tau_\theta, \tau_\psi)^T$ to rotor speeds.

Physical constraint

$\Omega_i^2 \geq 0$ always — rotors cannot reverse thrust!

Neglected Aerodynamic Effects

The simple quadratic model is a **first-order approximation**. Effects ignored at this level:

- **Blade flapping** — blades flex under load
- **Induced drag** — velocity-dependent resistance
- **Ground effect** — increased thrust near surfaces
- **Motor dynamics** — first-order lag of ESC+motor
- **Air density** — altitude and temperature dependence
- **Fuselage drag** — body drag at high speed

Rule of thumb

The quadratic model is valid for **near-hover** operation ($|\phi|, |\theta| < 20$, moderate speed).

Newton–Euler Formulation

The quadrotor is modelled as a **rigid body** of mass m with inertia tensor $I = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ (by symmetry).

Translational dynamics (world frame)

$$m\ddot{\xi} = R \begin{pmatrix} 0 \\ 0 \\ f \end{pmatrix} - mg e_3$$

$e_3 = (0, 0, 1)^T$ — gravity acts downward.

Rotational dynamics (body frame)

$$I \dot{\omega}^B = -\omega^B \times (I \omega^B) + \tau$$

$\tau = (\tau_\phi, \tau_\theta, \tau_\psi)^T$ — net torque in body frame.

Scalar Equations of Motion

Writing out all six scalar ODEs (in world / Euler-angle coordinates):

$$\ddot{x} = \frac{f}{m} (c_\psi s_\theta c_\phi + s_\psi s_\phi)$$

$$\ddot{y} = \frac{f}{m} (s_\psi s_\theta c_\phi - c_\psi s_\phi)$$

$$\ddot{z} = \frac{f}{m} c_\theta c_\phi - g$$

$$\ddot{\phi} = \dot{\theta} \dot{\psi} \frac{I_{yy} - I_{zz}}{I_{xx}} + \frac{\tau_\phi}{I_{xx}}$$

$$\ddot{\theta} = \dot{\phi} \dot{\psi} \frac{I_{zz} - I_{xx}}{I_{yy}} + \frac{\tau_\theta}{I_{yy}}$$

$$\ddot{\psi} = \dot{\phi} \dot{\theta} \frac{I_{xx} - I_{yy}}{I_{zz}} + \frac{\tau_\psi}{I_{zz}}$$

The cross-products $\dot{\alpha} \dot{\beta}(\cdot)$ are **gyroscopic** / **centrifugal** coupling terms.

Gyroscopic Effect of the Rotors

Each spinning rotor has angular momentum $h_i = J_r \Omega_i$ (J_r = rotor moment of inertia about its spin axis).

The **total rotor gyroscopic torque** adds to the rotational dynamics:

$$\tau_{gyro} = J_r \omega^B \times e_3 \Omega_{res}$$

where $\Omega_{res} = \Omega_1 - \Omega_2 + \Omega_3 - \Omega_4$ (signed sum).

At hover

$\Omega_{res} = 0$ by design (CW/CCW pairs cancel) $\Rightarrow$ no gyroscopic disturbance.

During maneuvers

$\Omega_{res} \neq 0$ when asymmetric thrust $\Rightarrow$ couples roll/pitch.

State-Space Representation

Define the full state vector:

$$x = (x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, p, q, r)^T \in \mathbb{R}^{12}$$

with inputs $u = (f, \tau_\phi, \tau_\theta, \tau_\psi)^T \in \mathbb{R}^4$.

The equations of motion take the compact form:

$$\dot{x} = f(x, u)$$

Summary of system properties

DOF	6 (position + orientation)
Inputs	4 (thrust + 3 torques)
State dim.	12
Linearity	Nonlinear
Actuation	Underactuated

Hover Equilibrium

At hover, the equilibrium point is:

$$x^* = 0_{12}, \quad u^* = (mg, 0, 0, 0)^T$$

(vehicle stationary, level, four rotors spinning at $\Omega^* = \sqrt{mg/(4k_f)}$).

Linearize $\dot{x} = f(x, u)$ around (x^*, u^*)

$$\delta \dot{x} = A \delta x + B \delta u$$

with $A = \partial f / \partial x|_*$ and $B = \partial f / \partial u|_*$.

Decoupled Linear Model

Near hover (small angles), the system **approximately decouples** into four independent subsystems:

Altitude (z)

$$\ddot{z} = \frac{\delta f}{m}$$

Double integrator in z .

Roll-y channel

$$\ddot{\phi} = \frac{\tau_{\phi}}{I_{xx}}, \quad \ddot{y} = g \delta \phi$$

Coupled: roll $\leftrightarrow$ lateral motion.

Yaw (ψ)

$$\ddot{\psi} = \frac{\tau_{\psi}}{I_{zz}}$$

Double integrator in ψ .

Pitch-x channel

$$\ddot{\theta} = \frac{\tau_{\theta}}{I_{yy}}, \quad \ddot{x} = -g \delta \theta$$

Stability Analysis of the Hover

Open-loop hover is unstable

All eigenvalues of A are on the imaginary axis or in the right half-plane (double integrators, no damping)
⇒ **cannot fly without feedback.**

Eigenvalue structure of the linearised model:

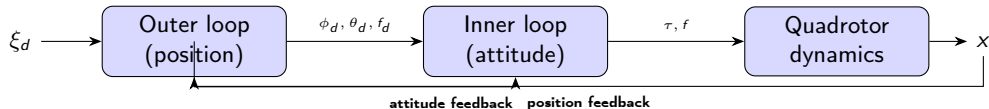
Subsystem	Eigenvalues	Stability
Altitude	0, 0	Marginally stable
Yaw	0, 0	Marginally stable
Roll-y	$0, 0, \pm \sqrt{g/I_{xx}}$	Unstable
Pitch-x	$0, 0, \pm \sqrt{g/I_{yy}}$	Unstable

The translational-rotational coupling creates unstable modes.

Cascade (Inner–Outer Loop) Structure

Standard architecture

Two nested control loops exploiting time-scale separation:



- **Outer loop** (slow): position error $\rightarrow$ desired attitude (ϕ_d, θ_d) and thrust f_d
- **Inner loop** (fast): attitude error $\rightarrow$ body torques τ

PD Control for Altitude and Attitude

Altitude controller

$$f = m(\ddot{z}_d + K_{d,z}(\dot{z}_d - \dot{z}) + K_{p,z}(z_d - z) + g)$$

Attitude controller (per axis)

$$\tau_\phi = I_{xx}(\ddot{\phi}_d + K_{d,\phi}(\dot{\phi}_d - p) + K_{p,\phi}(\phi_d - \phi))$$

$$\tau_\theta = I_{yy}(\ddot{\theta}_d + K_{d,\theta}(\dot{\theta}_d - q) + K_{p,\theta}(\theta_d - \theta))$$

$$\tau_\psi = I_{zz}(\ddot{\psi}_d + K_{d,\psi}(\dot{\psi}_d - r) + K_{p,\psi}(\psi_d - \psi))$$

These are **computed-torque** (feedback linearization) laws — the same structure as Ch. 5 for serial manipulators.

Desired Attitude from Position Error

The outer loop converts a position error into a **desired tilt**:

Virtual acceleration command

$$a_{cmd} = \ddot{\xi}_d + K_d(\dot{\xi}_d - \dot{\xi}) + K_p(\xi_d - \xi)$$

Desired thrust and attitude

$$f_d = m \|a_{cmd} + g e_3\|$$

$$\phi_d = \arcsin\left(\frac{m a_{cmd,x} s_{\psi_d} - m a_{cmd,y} c_{\psi_d}}{f_d}\right)$$

$$\theta_d = \arctan\left(\frac{m a_{cmd,x} c_{\psi_d} + m a_{cmd,y} s_{\psi_d}}{f_d + m a_{cmd,z}}\right)$$

Gain Tuning Guidelines

Stability requires $K_p, K_d > 0$. Critically damped response: $K_d = 2\sqrt{K_p}$.

Attitude loop (inner — tune first)

Parameter	Typical value	Effect
$K_{p,\phi}$	6–10	Roll stiffness
$K_{d,\phi}$	4–6	Roll damping
$K_{p,\theta}$	6–10	Pitch stiffness
$K_{d,\theta}$	4–6	Pitch damping

Position loop (outer — tune second)

Parameter	Typical value	Effect
$K_{p,z}$	2–4	Altitude stiffness
$K_{d,z}$	2–3	Altitude damping
$K_{p,xy}$	1–3	Horizontal stiffness
$K_{d,xy}$	1–2	Horizontal damping

RK4 Integration of the Equations of Motion

The 12-state ODE $\dot{x} = f(x, u)$ is integrated with **4th-order Runge–Kutta** (same routine as `dynamics.simulate()` in the course library):

RK4 step at time t_k

$$k_1 = f(x_k, u_k)$$

$$k_2 = f(x_k + \frac{h}{2} k_1, u_{k+\frac{1}{2}})$$

$$k_3 = f(x_k + \frac{h}{2} k_2, u_{k+\frac{1}{2}})$$

$$k_4 = f(x_k + h k_3, u_{k+1})$$

$$x_{k+1} = x_k + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

h Integration step — typically 1–10 ms

Error $\mathcal{O}(h^5)$ per step, $\mathcal{O}(h^4)$ global

Simulation Parameters — Crazyflie 2.x

Symbol	Value	Description
m	0.027 kg	Total mass
l	0.046 m	Arm length
I_{xx}	$1.6 \times 10^{-5} \text{ kg}\cdot\text{m}^2$	Roll inertia
I_{yy}	$1.7 \times 10^{-5} \text{ kg}\cdot\text{m}^2$	Pitch inertia
I_{zz}	$2.9 \times 10^{-5} \text{ kg}\cdot\text{m}^2$	Yaw inertia
k_f	$3.16 \times 10^{-10} \text{ N/rpm}^2$	Thrust coeff.
k_m	$7.94 \times 10^{-12} \text{ N}\cdot\text{m/rpm}^2$	Drag coeff.
g	9.81 m/s ²	Gravity
h	1 ms	RK4 step

Scenario: Vertical Climb + Lateral Step

A typical test trajectory:

Phase 1 — Vertical takeoff (0–2 s)

Command $z_d(t) = 0.5$ m step from ground.

Expect: overshoot $< 10\%$, settling time < 2 s.

Phase 2 — Lateral step (2–5 s)

Command $x_d = 1$ m while holding $z_d = 0.5$ m.

Outer loop generates $\theta_d \neq 0 \Rightarrow$ roll couples with y .

Phase 3 — Return to hover (5–8 s)

Command $x_d = 0$ m.

Verify symmetry of the response.

Identifying k_f and k_m

Thrust stand test

- 1 Mount single rotor on load cell
- 2 Sweep Ω from 0 to max RPM
- 3 Fit $F = k_f \Omega^2$ via least squares
- 4 Repeat for drag torque measurement

Hover test (indirect)

At hover: $f^* = mg$

$$\Omega^* = \sqrt{\frac{mg}{4k_f}}$$

Measure Ω^* from ESC telemetry $\Rightarrow$ solve for k_f .

Typical k_f/k_m ratio

$k_m/k_f \approx 10^{-2}$ (depends on blade pitch and diameter).

Identifying the Inertia Tensor I

CAD method

Import 3D model into SolidWorks / Fusion 360 with correct material densities $\Rightarrow$ automatic I computation.

Pros: fast, no experiment.

Cons: ignores electronics, wiring.

Bifilar pendulum

Suspend UAV from two parallel strings of length L_s and radius r . Measure oscillation period T_p :

$$I_{zz} = \frac{m g r^2 T_p^2}{4\pi^2 L_s}$$

Rotate for I_{xx} , I_{yy} .

Geometric Control on $SO(3)$

PD/PID control works near hover, but breaks at large angles. **Geometric control** avoids Euler singularities by working directly on the Lie group $SO(3)$:

Attitude error on $SO(3)$

$$e_R = \frac{1}{2} (R_d^T R - R^T R_d)^\vee$$

$(\cdot)^\vee$ = vex operator (extracts axial vector from skew-symmetric matrix).

Control law (Lee et al., 2010)

$$\tau = -K_R e_R - K_\omega e_\omega + \omega \times I \omega$$

Advantage

Globally valid — handles flips, aggressive acrobatics without singularity.

Model Predictive Control (MPC)

Idea

At each time step, solve an **optimal control problem** over a finite horizon N :

$$\min_{u_0, \dots, u_{N-1}} \sum_{k=0}^{N-1} (x_k - x_k^*)^T Q (x_k - x_k^*) + u_k^T R u_k$$

subject to dynamics, actuator constraints, and safety constraints.

Advantages

- Handles constraints explicitly
- Predictive $\Rightarrow$ less overshoot
- Can include obstacle avoidance

Challenges

- Computationally expensive
- Requires fast QP solver (OSQP, acados)
- Model accuracy critical

What we have covered

- 1 **Architecture:** $\times$ and $+$ configurations, CW/CCW alternation
- 2 **Kinematics:** world/body frames, ZYX Euler angles, rotation matrix R
- 3 **Aerodynamics:** $f_i = k_f \Omega_i^2$, allocation matrix Γ
- 4 **Dynamics:** Newton–Euler, 12-state ODE, underactuation
- 5 **Linearization:** hover equilibrium, decoupled subsystems, instability
- 6 **Control:** cascade PD, computed-torque structure, gain tuning
- 7 **Simulation:** RK4 integration, Crazyflie parameters
- 8 **Identification:** thrust-stand, bifilar pendulum
- 9 **Advanced:** geometric control on $SO(3)$, MPC

References

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